

- Paso 1:
- Paso 2:
- Paso 3:
- Paso 4:
- Paso 5:
- Paso 6:
- Paso 7:
- Paso 8:
- Paso 9:
- Paso 10:
- Paso 11:
- Paso 12: Comprobaciones

Paso 1:

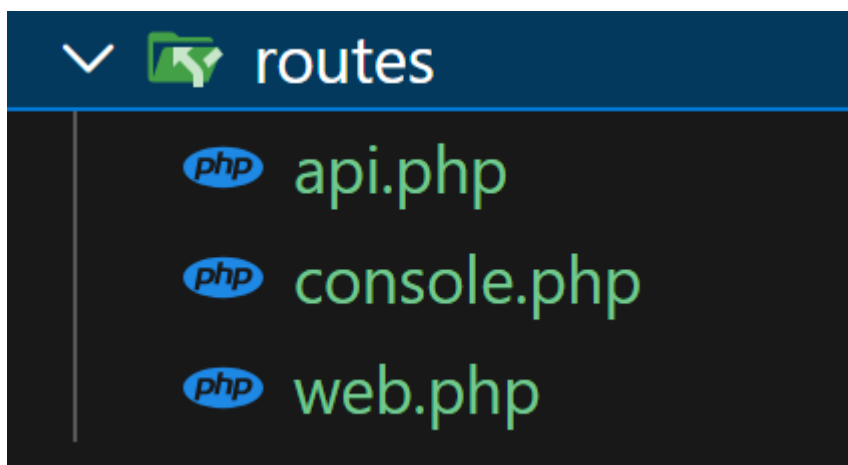
Crear el proyecto

```
PS C:\xampp\htdocs\DWES\Dwes_Maria\UD7APIREST\ProyectoFinal> composer create-project laravel/laravel api-animales
- Installing symfony/process (v7.4.5): Extracting archive
```

Paso 2:

Crear el soporte para API (Sanctum) y crear routes/api

```
PS C:\xampp\htdocs\DWES\Dwes_Maria\UD7APIREST\ProyectoFinal\api-animales> php artisan install:api
./composer.json has been updated
Running composer update laravel/sanctum
```



Paso 3:

Creamos el modelo Animal con todos los archivos auxiliares, migración, factory y seeder

```
PS C:\xampp\htdocs\DWES\Dwes_Maria\UD7APIREST\ProyectoFinal\api-animales> php artisan make:model Animal -mfs
```

```
INFO Model [C:\xampp\htdocs\DWES\Dwes_Maria\UD7APIREST\ProyectoFinal\api-animales\app\Models\Animal.php] created successfully.
INFO Factory [C:\xampp\htdocs\DWES\Dwes_Maria\UD7APIREST\ProyectoFinal\api-animales\database\factories\AnimalFactory.php] created successfully.
INFO Migration [C:\xampp\htdocs\DWES\Dwes_Maria\UD7APIREST\ProyectoFinal\api-animales\database\Migrations\2026_02_22_115131_create_animals_table.php] created successfully.
INFO Seeder [C:\xampp\htdocs\DWES\Dwes_Maria\UD7APIREST\ProyectoFinal\api-animales\database\seeders\AnimalSeeder.php] created successfully.
```

Modificamos la migración

```
public function up(): void
{
    Schema::create('animals', function (Blueprint $table): void {
        $table->id();
        $table->string('name');
        $table->string('species');
        $table->integer('age');
        $table->string('owner_email');
        $table->string('image_path')->nullable();
        $table->timestamps();
    });
}
```

Modificamos el modelo

```

Dwes_Maria > UD7APIREST > ProyectoFinal > api-animales > app > Models > Animal.php > PHP >
1  <?php
2
3  namespace App\Models;
4
5  use Illuminate\Database\Eloquent\Factories\HasFactory;
6  use Illuminate\Database\Eloquent\Model;
7
8  class Animal extends Model
9  {
10     use HasFactory;
11
12     protected $fillable = [
13         'name',
14         'species',
15         'age',
16         'owner_email',
17         'image_path'
18     ];
19 }

```

Modificamos el factory

```

Dwes_Maria > UD7APIREST > ProyectoFinal > api-animales > database > factories > AnimalFactory.php > PHP >
5  use Illuminate\Database\Eloquent\Factories\Factory;
6
7  /**
8   * @extends \Illuminate\Database\Eloquent\Factories\Factory<\App\Models\Animal>
9   */
10 class AnimalFactory extends Factory
11 {
12     /**
13      * Define the model's default state.
14      *
15      * @return array<string, mixed>
16      */
17     public function definition(): array
18     {
19         return [
20             'name' => $this->faker->firstName(),
21             'species' => $this->faker->randomElement(['Perro', 'Gato', 'Loro', 'Hamster']),
22             'age' => $this->faker->numberBetween(1, 15),
23             'owner_email' => $this->faker->safeEmail(),
24         ];
25     }
26 }

```

Modificamos el seeder

```

Dwes_Maria > UD7APIREST > ProyectoFinal > api-animales > database > seeders >  Animal
1   <?php
2
3   namespace Database\Seeders;
4
5   use App\Models\Animal;
6   use Illuminate\Database\Seeder;
7
8   class AnimalSeeder extends Seeder
9   {
10
11       public function run(): void
12       {
13           Animal::factory()->count(10)->create();
14       }
15

```

Llamamos al seeder en database seeder

```

Dwes_Maria > UD7APIREST > ProyectoFinal > api-animales > database > seeders >  DatabaseSeeder.php >  DatabaseSeeder
2
3   namespace Database\Seeders;
4
5   use App\Models\User;
6   use App\Models\Animal;
7   use Illuminate\Database\Console\Seeds\WithoutModelEvents;
8   use Illuminate\Database\Seeder;
9   use Illuminate\Support\Facades\Hash;
10
11   0 references | 0 implementations
12   class DatabaseSeeder extends Seeder
13   {
14       use WithoutModelEvents;
15
16       /**
17        * Seed the application's database.
18        */
19       0 references | 0 overrides
20       public function run(): void
21       {
22           // Creamos el usuario administrador para las pruebas de la API
23           User::factory()->create([
24               'name' => 'Admin',
25               'email' => 'admin@admin.com',
26               'password' => Hash::make(value: '123456'), // Usamos Hash por seguridad
27           ]);
28
29           // Llamamos al seeder de animales que creamos antes
30           $this->call(class: AnimalSeeder::class);
31       }
32   }

```

Paso 4:

Ejecutamos las migraciones

```
PS C:\xampp\htdocs\DWES\Dwes_Maria\UD7APIREST\ProyectoFinal\api-animales> php artisan migrate --seed
```

Paso 5:

Creamos la validación:

```
PS C:\xampp\htdocs\DWES\Dwes_Maria\UD7APIREST\ProyectoFinal\api-animales> php artisan make:request AnimalRequest
```

```
INFO Request [C:\xampp\htdocs\DWES\Dwes_Maria\UD7APIREST\ProyectoFinal\api-animales\app\Http\Requests\AnimalRequest.php] created successfully.
```

```
Dwes_Maria > UD7APIREST > ProyectoFinal > api-animales > app > Http > Requests > php AnimalRequest.php > PH
```

```
1  <?php
2
3  namespace App\Http\Requests;
4
5  use Illuminate\Foundation\Http\FormRequest;
6
7  0 references | 0 implementations
8  class AnimalRequest extends FormRequest
9  {
10     0 references | 0 overrides
11     public function authorize(): bool
12     {
13         return true;
14     }
15
16     0 references | 0 overrides
17     public function rules(): array
18     {
19         return [
20             'name' => 'required|string|max:255',
21             'species' => 'required|string|max:100',
22             'age' => 'required|integer|min:0',
23             'owner_email' => 'required|email',
24             'image' => 'nullable|image|mimes:jpg,png,jpeg|max:2048'
```

Paso 6:

Creamos el transformador de datos

```
PS C:\xampp\htdocs\DWES\Dwes_Maria\UD7APIREST\ProyectoFinal\api-animales> php artisan make:resource AnimalResource
```

```
INFO Resource [C:\xampp\htdocs\DWES\Dwes_Maria\UD7APIREST\ProyectoFinal\api-animales\app\Http\Resources\AnimalResource.php] created successfully.
```

```
<?php
```

```
namespace App\Http\Resources;
```

```
use Illuminate\Http\Request;
```

```
use Illuminate\Http\Resources\Json\JsonResource;
```

```
0 references | 0 implementations
```

```
class AnimalResource extends JsonResource
```

```
{
```

```
0 references | 0 overrides | prototype
```

```
public function toArray(Request $request): array
```

```
{
```

```
    return [
```

```
        'id' => $this->id,
```

```
        'nombre_animal' => $this->name, // Ejemplo de renombrar campo
```

```
        'especie' => $this->species,
```

```
        'edad' => $this->age,
```

```
        'contacto_dueno' => $this->owner_email,
```

```
        'foto' => $this->image_path ? asset(path: '..storage/' . $this->image_path) : null,
```

```
        'fecha_registro' => $this->created_at->format('d-m-Y')
```

```
    ];
```

```
}
```

```
}
```

Paso 7:

Creamos el controller, si usamos --api no nos creará ni create ni edit

```
... php artisan make:controller Api/V1/AnimalController --api
```

```
PS C:\xampp\htdocs\DWES\Dwes_Maria\UD7APIREST\ProyectoFinal\api-animales>
```

```
INFO Controller [C:\xampp\htdocs\DWES\Dwes_Maria\UD7APIREST\ProyectoFinal\api-animales\app\Http\Controllers\Api\V1\AnimalController.php] created successfully.
```

```
PS C:\xampp\htdocs\DWES\Dwes_Maria\UD7APIREST\ProyectoFinal\api-animales>
```

```
1  <?php
2
3  namespace App\Http\Controllers\Api\V1;
4
5  use App\Http\Controllers\Controller;
6  use App\Models\Animal;
7  use App\Http\Requests\AnimalRequest;
8  use App\Http\Resources\AnimalResource;
9  use Illuminate\Http\JsonResponse;
10 use Illuminate\Http\Resources\Json\JsonResource;
11
12 class AnimalController extends Controller
13 {
14     // Mostrar todos los animales
15
16     public function index(): JsonResponse
17     {
18         return AnimalResource::collection(resource: Animal::all());
19     }
20
21     // Crear un animal
22
23     public function store(AnimalRequest $request): JsonResponse
24     {
25         $data = $request->validated();
26
27         // Simulación de tratamiento de imagen (lo profundizaremos luego)
28         if ($request->hasFile(key: 'image')) {
29             $data['image_path'] = $request->file(key: 'image')->store(path: 'animals', options: 'public');
30         }
31
32         $animal = Animal::create(attributes: $data);
33         return response()->json(data: [
34             'success' => true,
35             'data' => new AnimalResource(resource: $animal)
36         ], status: 201);
37     }
38
39     // Mostrar uno solo
40
41     public function show(Animal $animal): AnimalResource
42     {
43         return new AnimalResource(resource: $animal);
44     }
45
46     // Actualizar
47
48     public function update(AnimalRequest $request, Animal $animal): JsonResponse
49     {
50         $animal->update(attributes: $request->validated());
51         return response()->json(data: [
52             'success' => true,
53             'data' => new AnimalResource(resource: $animal)
54         ], status: 200);
55     }
56
57     // Eliminar
58
59     public function destroy(Animal $animal): JsonResponse
60     {
61         $animal->delete();
62         return response()->json(data: ['success' => true], status: 204);
63     }
64 }
```

Paso 8:

Modificamos las rutas

Dwes_Maria > UD7APIREST > ProyectoFinal > api-animales > routes >  api.php

```
1  <?php
2
3  use App\Http\Controllers\Api\V1\AnimalController;
4  use Illuminate\Support\Facades\Route;
5
6  Route::prefix(prefix: 'v1')->group(callback: function (): void {
7  |   Route::apiResource(name: 'animals', controller: AnimalController::class);
8  | });
```

Paso 9:

Modificamos el userModel

Dwes_Maria > UD7APIREST > ProyectoFinal > api-animales > app > Models >  User.php > PHP >  User >  hidden

```
1  <?php
2
3  namespace App\Models;
4
5  use Illuminate\Database\Eloquent\Factories\HasFactory;
6  use Illuminate\Foundation\Auth\User as Authenticatable;
7  use Illuminate\Notifications\Notifiable;
8  use Laravel\Sanctum\HasApiTokens;
9
10 76 references | 0 implementations
11 class User extends Authenticatable
12 {
13     use HasApiTokens, HasFactory, Notifiable;
14
15     0 references
16     protected $fillable = [
17         'name',
18         'email',
19         'password',
20     ];
21
22     0 references
23     protected $hidden = [
24         'password',
25         'remember_token',
26     ];
27 }
```

Paso 10:

Creamos el authController

PS C:\xampp\htdocs\DWES\Dwes_Maria\UD7APIREST\ProyectoFinal\api-animales> php artisan make:controller Api/V1/AuthController

INFO Controller [C:\xampp\htdocs\DWES\Dwes_Maria\UD7APIREST\ProyectoFinal\api-animales\app\Http\Controllers\Api\V1\AuthController.php] created successfully.


```

es_Maria > UD7APIREST > ProyectoFinal > api-animales > app > Http > Controllers > Api > V1 > AuthController.php > PHP > AuthController >
2
3 namespace App\Http\Controllers\Api\V1;
4
5 use App\Http\Controllers\Controller;
6 use App\Models\User;
7 use Illuminate\Http\Request;
8 use Illuminate\Support\Facades\Hash;
9 use Illuminate\Validation\ValidationException;
10
11 0 references | 0 implementations
12 class AuthController extends Controller
13 {
14     0 references | 0 overrides
15     public function login(Request $request): JsonResponse
16     {
17         $request->validate(rules: [
18             'email' => 'required|email',
19             'password' => 'required',
20         ]);
21
22         $user = User::where(column: 'email', operator: $request->email)->first();
23
24         if (! $user || ! Hash::check(value: $request->password, hashedValue: $user->password)) {
25             return response()->json(data: ['message' => 'Credenciales incorrectas'], status: 401);
26         }
27
28         // Creamos el token (este es el "secreto" que usará el Front)
29         $token = $user->createToken(name: 'auth_token')->plainTextToken;
30
31         return response()->json(data: [
32             'access_token' => $token,
33             'token_type' => 'Bearer',
34             'user' => $user
35         ]);
36     }
37
38     0 references | 0 overrides
39     public function logout(Request $request): JsonResponse
40     {
41         $request->user()->currentAccessToken()->delete();
42         return response()->json(data: ['message' => 'Sesión cerrada']);
43     }
44 }

```

Paso 11:

Volvemos a modificar el archivo de las rutas, añadiendo las rutas publicas y protegidas

Dwes_Maria > UD7APIREST > ProyectoFinal > api-animales > routes >  api.php

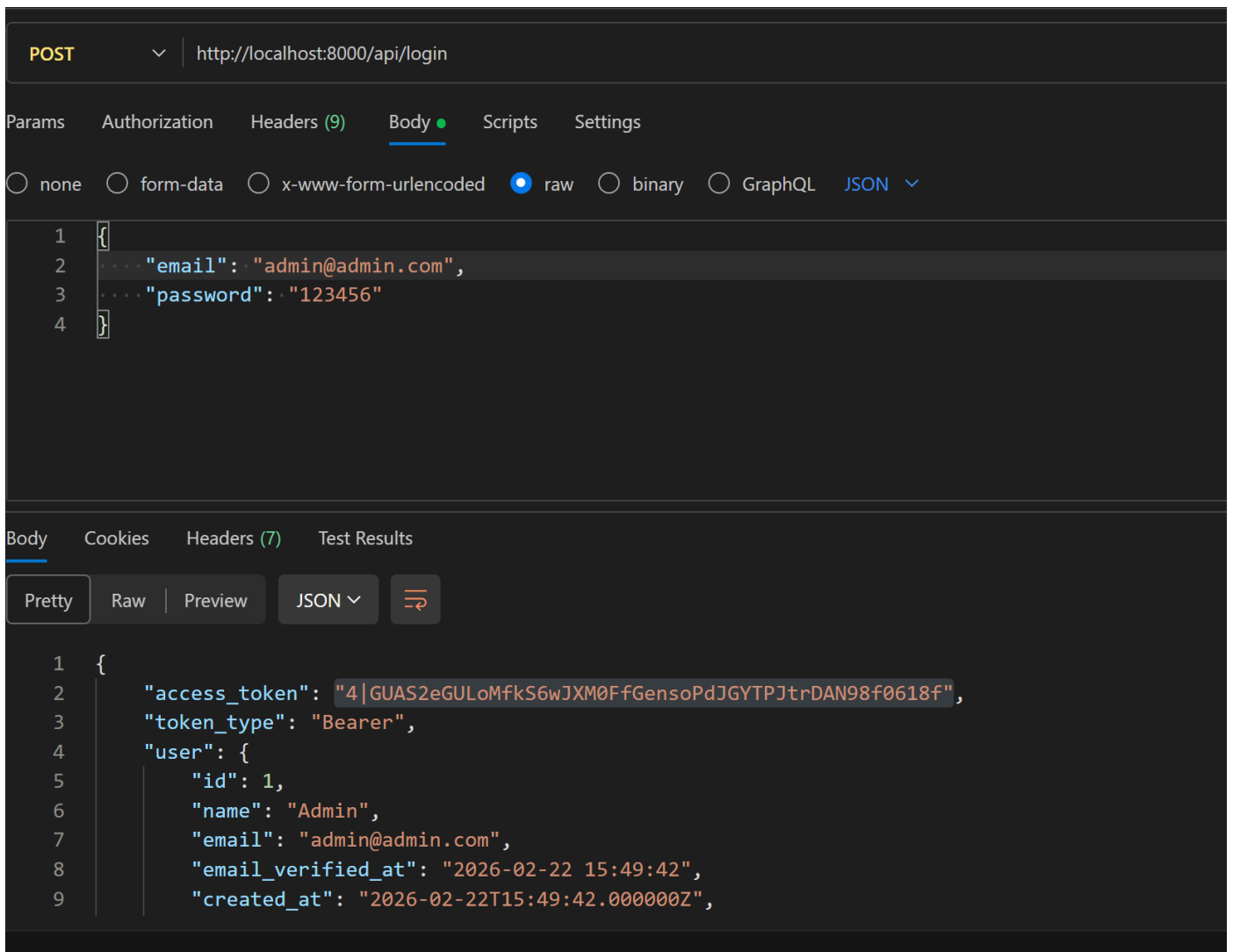
```
1  <?php
2
3  use App\Http\Controllers\Api\V1\AnimalController;
4  use App\Http\Controllers\Api\V1\AuthController;
5  use Illuminate\Support\Facades\Route;
6
7  // Rutas públicas de Auth
8  Route::post(uri: '/login', action: [AuthController::class, 'login']);
9
10 Route::prefix(prefix: 'v1')->group(callback: function (): void {
11
12     // Ruta pública para ver animales (INDEX y SHOW)
13     Route::get(uri: 'animals', action: [AnimalController::class, 'index']);
14     Route::get(uri: 'animals/{animal}', action: [AnimalController::class, 'show']);
15
16     // Rutas protegidas (STORE, UPDATE, DESTROY)
17     Route::middleware(middleware: 'auth:sanctum')->group(callback: function (): void {
18         Route::post(uri: 'animals', action: [AnimalController::class, 'store']);
19         Route::put(uri: 'animals/{animal}', action: [AnimalController::class, 'update']);
20         Route::delete(uri: 'animals/{animal}', action: [AnimalController::class, 'destroy']);
21         Route::post(uri: '/logout', action: [AuthController::class, 'logout']);
22     });
23 });
```

Paso 12: Comprobaciones

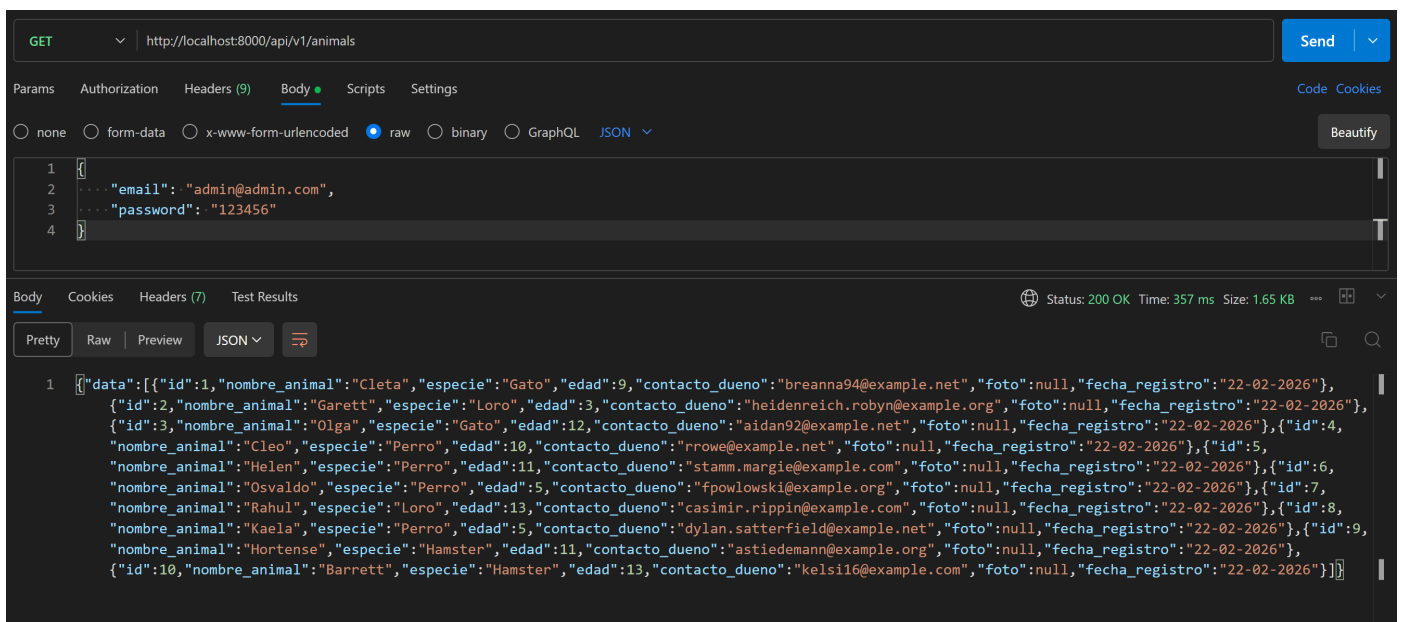
Las comprobaciones las he realizado en postman, al probar el login, en el body vemos que nos devuelve el token de acceso bearer

TOKEN

"4|GUAS2eGULoMfkS6wJXM0FfGensoPdJGYTPJtrDAN98f0618f"



La siguiente comprobación que haremos será la de devolver todos los animales



Probamos ahora a buscar y obtener la información de un solo animal por su id, en este caso el 3 y nos devuelve correctamente la información

GET

⌵

http://localhost:8000/api/v1/animals/3

Params

Authorization

Headers (7)

Body

Scripts

Settings

Query Params

	Key
	Key

Body

Cookies

Headers (7)

Test Results

Pretty

Raw

Preview

JSON ⌵

⌵

```
2      "data": {
3          "id": 3,
4          "nombre_animal": "Olga",
5          "especie": "Gato",
6          "edad": 12,
7          "contacto_dueno": "aidan92@example.net",
8          "foto": null,
9          "fecha_registro": "22-02-2026"
10     }
```

Añadimos ahora con animal para ellos deberemos de usar el token ya que sino no nos dejara

HTTP

http://localhost:8000/api/v1/animals

POST

http://localhost:8000/api/v1/animals

Params

Authorization

Headers (10)

Body

Scripts

Settings

Type

Bearer Token

Token

GUAS2eGULoMfkS6wJXM0FfGensoPdJGYT...

The authorization header will be automatically generated when you send the request. Learn more about [authorization](#)

Body

Cookies

Headers (7)

Test Results

Status: 201 Created

Pretty

Raw

Preview

JSON

```
1 {
2   "success": true,
3   "data": {
4     "id": 11,
5     "nombre_animal": "Rex",
6     "especie": "Perro",
7     "edad": 3,
8     "contacto_dueno": "juan@example.com",
9     "foto": null,
```

11	11	Rex	Perro	3	juan@example.com	NULL	2026-02-22 16:...	201
----	----	-----	-------	---	------------------	------	-------------------	-----

De la misma forma podemos actualizar un animal



http://localhost:8000/api/v1/animals/11

PUT



http://localhost:8000/api/v1/animals/11

Params

Authorization ●

Headers (10)

Body ●

Scripts

Settings

☐ none

☐ form-data

☐ x-www-form-urlencoded

☒ raw

☐ binary

☐ GraphQL

JSON

```
1 {
2   .... "name": "Rex",
3   .... "species": "Gato",
4   .... "age": 3,
5   .... "owner_email": "juan@example.com"
6 }
```

Body

Cookies

Headers (7)

Test Results

Pretty

Raw

Preview

JSON ▾



```
1 {
2   "success": true,
3   "data": {
4     "id": 11,
5     "nombre_animal": "Rex",
6     "especie": "Gato",
7     "edad": 3,
8     "contacto_dueno": "juan@example.com",
9     "foto": null,
```

11

11 Rex

Gato

3 juan@example.com

NULL

2026-02-22 16:...

20

Y por último eliminamos a un animal

http://localhost:8000/api/v1/animals/11

SaveNo Environment

DELETE

http://localhost:8000/api/v1/animals/11

Send

Params

Authorization

Headers (10)

Body

Scripts

Settings

Code Co

none

form-data

x-www-form-urlencoded

raw

binary

GraphQL

JSON

Bea

```
1 {
2   ... "name": "Rex",
3   ... "species": "Gatohttp://localhost:8000/api/v1/animals/11",
4   ... "age": 3,
5   ... "owner_email": "juan@example.com"
6 }
```

Body

Cookies

Headers (6)

Test Results

204 No Content

Time: 235 ms

Size: 197 B

...

204 No Content

The server successfully processed the request, but is not returning any content.

Pretty

Raw

Preview

Text

1

	id	name	species	age	owner_email	image_p...	created_at	up
	Filter	Filter	Filter	Filter	Filter...	Filter	Filter...	F
1	1	Cleta	Gato	9	breanna94@example.net	NULL	2026-02-22 15:...	20
2	2	Garett	Loro	3	heidenreich.robyn@example.org	NULL	2026-02-22 15:...	20
3	3	Olga	Gato	12	aidan92@example.net	NULL	2026-02-22 15:...	20
4	4	Cleo	Perro	10	rrowe@example.net	NULL	2026-02-22 15:...	20
5	5	Helen	Perro	11	stamm.margie@example.com	NULL	2026-02-22 15:...	20
6	6	Osvaldo	Perro	5	fpowlowski@example.org	NULL	2026-02-22 15:...	20
7	7	Rahul	Loro	13	casimir.rippin@example.com	NULL	2026-02-22 15:...	20
8	8	Kaela	Perro	5	dylan.satterfield@example.net	NULL	2026-02-22 15:...	20
9	9	Hortense	Hamster	11	astiedemann@example.org	NULL	2026-02-22 15:...	20
10	10	Barrett	Hamster	13	kelsi16@example.com	NULL	2026-02-22 15:...	20
+	11							